

Education

Monash University

Department of Econometrics and
Business Statistics

Doctor of Philosophy

June 2022 - present

- Thesis title: "Investigations of Flexible Jump Modelling and Systematic Risk Implied by Option Information"
- Commenced in the University of Melbourne (June 2022) and transferred to Monash with main supervisor (from Feb 2023).

The University of New South Wales

Department of Finance

Visiting PhD

Oct, 2024 - Dec, 2024

- Hosted by Andrew Patton

University of Sydney

Department of Economics

Graduate Certificate in Economics Analysis

Aug 2021 - Mar 2022

- Major in Economics
- WAM: 86.5

Monash University

Monash Business School

Bachelor of Commerce (1st Class Honors)

Feb 2017 - Dec 2020

- Major in Finance and Econometrics
- Graduated with High Distinction
- GPA: 3.9/4.0

Job Market Paper

Learning Financial Jumps with Infinite Gaussian Mixture

Monash University & University of Melbourne

June 2022 - Present

- **Summary:** A Bayesian nonparametric method using Dirichlet process mixture to model financial price jumps size distribution and improve the predictive distribution of returns.
- [Link to the paper](#)

Working Papers

Downside and Upside Systematic Risk Measure Implied by Option Information

Monash University

Nov 2024 - Present

- **Summary:** An innovative method of constructing the option implied downside and upside betas for systematic risk measures through option implied truncated moments.

Optimal Probabilistic Forecasts for Risk Management

Monash University

Mar 2020 - Mar 2023

- **Summary:** This paper studies how choosing tail-focused scoring rules to calibrate forecast distributions affects risk management performance in practice. It shows that tail-focused calibration improves Value-at-Risk and expected shortfall accuracy, and also leads to better hedging outcomes when forecasting the VIX for strategies using VIX futures.
- [Link to the paper](#)

Conference Presentations

Bayes on the Beach

University of Wollongong, Australia

February, 2026

- Accepted

World Congress of the Econometric Society

Seoul, South Korea

August, 2025

The European Seminar on Bayesian Econometrics (ESOB 2025)

Melbourne Business School, Australia

August, 2025

- Poster presentation

The 8th International Conference on Econometrics and Statistics (EcoSta 2025) Waseda University, Japan Invited Talk	August, 2025
Annual Conference of the International Association for Applied Econometrics University of Turin, Italy	June, 2025
The 19th International Symposium on Econometric Theory and Applications: SETA 2025 University of Macau, China	June, 2025
	Accepted
Econometric Society Australian Meetings (ESAM) Monash University, Australia	December, 2024
Time Series and Forecasting Symposium University of Sydney, Australia Best Poster Runner-up	November, 2024

Work Experience

Monash Business School Teaching Associate - Delivered tutorials across undergraduate and postgraduate units, supporting students to interpret quantitative results and apply modelling tools to real datasets. Demonstrated reliable delivery under tight timelines and strong ability to communicate technical methods to non-technical audiences; - Recognised with Excellent Teaching Team Awards (Data Analysis in Business, 2024; Business Forecasting, 2025) and consistently achieved top-tier student evaluation outcomes. - Demonstrated strong planning and delivery skills through timely feedback during peak teaching periods while maintaining research progress.	Melbourne, Australia
	Feb 2023 - present
Monash Business School Research Assistant - Conducted empirical analysis on high-frequency bond market data to quantify market behaviour and assess the impact of regulatory and policy-related changes; - Built clean, reproducible workflows for data preparation and diagnostics, improving reliability of outputs across large datasets. - Delivered timely statistical modelling and synthesised quantitative findings into clear summaries to support decision-making.	Melbourne, Australia
	Aug 2023 - Oct 2024
Melbourne Business School Research Assistant - Automated large-scale high-frequency equity data retrieval and processing pipelines using Linux language, Python and R for millisecond return of 500 stocks, improving efficiency and reproducibility of dataset construction; - Cleaned and validated high-frequency time-series data, implemented quality checks, and produced analysis-ready datasets to support modelling and forecasting tasks; - Collaborated with academic stakeholders to translate project requirements into structured datasets and deliverables.	Melbourne, Australia
	Aug 2022 - Jan 2023
Liaowang Think Tank - Xinhua News Agency Research Analyst - Collaborated with small teams and contributed to policy-focused research projects by forming hypotheses, conducting macroeconomic analysis, and synthesising results into practical recommendations; - Produced written reports and presentation-ready materials for non-technical audiences, strengthening stakeholder communication and clarity of insights.	Beijing, China
	July 2021 - Oct 2021

Research Grants

Prestigious International Conference Award Monash Business School, 2025	2,300 AUD
CFM/SoFiE Scholarship Capital Fund Management, The Society for Financial Econometrics, 2025	Conference Accommodation Support
Departmental Funding for External Research Experience (Visiting Scholar Support) Department of Economics and Business Statistics at Monash University, 2024	6,400 AUD

Honors & Awards

2024	Excellent Teaching Team , Monash Business School
2024	Best Poster Runner Up , Time Series and Forecasting Symposium, University of Sydney
2023	Co-funded Monash Graduate Scholarship , Monash University
2023	Monash Business School Top-Up Scholarship , Monash Business School
2022	Faculty of Business and Economics Graduate Research Scholarship , University of Melbourne
2021	2020 Dean's Honor (First Class) , Monash Business School
2020	Econometrics Memorial Scholarship , Department of Econometrics and Business Statistics, Monash
2020	Meritorious Winner , 2020 Interdisciplinary contest in Modelling

Skills

Programming R, MATLAB, VBA, Python, Linux, Git, Excel

Referees

Ole Maneesoonthorn	Ole.Maneesoonthorn@monash.edu
Yong Song	yong.song@unimelb.edu.au
Wei Wei	Wei.Wei2@monash.edu